LESSON GUIDE

Technology Plus

Unit 6: Database

5.1.2 – Digital Products and Business Decisions

Lesson Overview:

In this lesson, students will learn why businesses use data to inform their decisions, improve business processes, increase revenue; and how to evaluate a business scenario and determine how the business can use data to their benefit.

Students will:

- Explain why companies use data to inform their business decisions.
- Understand how using data improves business processes and increases revenue.
- Evaluate business scenarios and develop a plan to utilize data for the businesses' benefit.

Guiding Question: How and why do companies use data to inform their business decisions?

Suggested Grade Levels: 8-12

Technology Needed: No

Approximate Time Required: 50-75 minutes

Standards:

CompTIA Tech+ FC0-U71 Objective

- 5.1 - Explain the value of data and information.
    - Data-driven business decisions
        - Data capture and collection
        - Data correlation
        - Meaningful reporting
    - Data monetization
    - Data analytics
    - Big Data

This content is based upon work supported by the US Department of Homeland Security's Cybersecurity & Infrastructure Security Agency under the Cybersecurity Education Training and Assistance Program (CETAP).

Lesson Background

Background Information:

Data is an asset to our society, it can provide economic value, competitive advantage, operational efficiency, and manage risks. Businesses utilize data to gain a strategic advantage and make as much money as possible. As businesses grow, adapt, and expand they are constantly capturing, correlating, and analyzing data to implement changes that would have a positive impact on their outcomes.

Materials Needed:

Materials per Class	• Lesson Guide • Lesson Slides 5.1.2 – Digital Products and Business Decisions • Activity Slides 5.1.2 – Data in Business
Materials per Student	• Student Notes 5.1.2 – Digital Products and Business Decisions • Student Handout 5.1.2 – Digital Products and Business Decisions • Assessment 5.1.2 – Digital Products and Business Decisions (optional)

Lesson Background

Cyber Terms:

Term	Definition
data capture	gathering raw data from various sources, which can include internal systems, customer interactions, external databases, and more
data correlation	the process of identifying relationships and patterns between different data sets
meaningful reporting	the process of presenting data and analysis that is clear, actionable, and aligned with business objectives
data monetization	the process of generating economic value from data
data analytics	the science of examining raw data to draw conclusions about that information
big data	extremely large amounts of information captured in digital format used with computing systems to identify trends and relationships

5.1.2 – Digital Products and Business Decisions

Guiding Question: How and why do companies use data to inform their business decisions?

Lesson Launch (5 minutes)

Why Data?

- Remind students they previously learned that data was an asset.
- Introduce to students how and why businesses can use data as an asset to make decisions.

Can You Data That? (30-45 minutes)

- Distribute the Student Handout 5.1.2 - Digital Products and Business Decisions and have students fill in the answers during the lesson.
- As tools and technologies are mentioned throughout the lesson, feel free to pull them up in a web browser to show to students.

Data-driven Business Decisions

- Introduce ***data capture*** and collection, discuss methods, tools, and best practices; elaborate as needed.
- Introduce ***data correlation***, discuss methods and applications, elaborate on how data correlation positively affects businesses.
- Introduce ***meaningful reporting***, discuss key components and best practices, elaborate on the importance of meaningful reporting and accurate data.

Data Monetization

- Introduce ***data monetization*** and its two strategies.
- Discuss direct monetization and indirect monetization, elaborate on the difference between the two strategies and discuss examples in detail.
- Discuss methods and considerations of data monetization, elaborate as needed.

Data Analytics

- Introduce ***data analytics*** and discuss the six-step process in detail, focusing on the purpose of each step.
- Discuss each type of data analytics, emphasizing the examples.
- Discuss the tools and technologies of data analytics, focus on the use of each.

- Discuss the real-world applications of data analytics, elaborate as needed.

Big Data

- Introduce ***big data*** and its characteristics, emphasize that traditional data processing tools are not effective in handling big data, elaborate as needed.
- Discuss big data technologies, elaborate as needed.
- Discuss real-world applications of big data, elaborate as needed.
- Discuss the challenges of big data, elaborate as needed.

Activity: Data in Business (10-15 minutes)

- Display Activity Slides 5.1.2 – Data in Business, have students write their answers for each scenario at the bottom of their handout.
- Read each scenario aloud or let students read independently. Provide students with time to write their answers.
- Some scenarios can have a variety of answers, all should be reasonable.

1. A business wants to receive monetary gain from data.

   First, the business can capture/collect data from customers by using transactional, survey, or social media data. Then, the business can use direct monetization to sell to data brokers or in the marketplaces, license the data to other companies, or offer access to data through a subscription. Accept all reasonable answers.

2. A business wants to retain and attract new customers.

   First, the business should capture/collect data through surveys and social media. The business can use data correlation to identify relationships and patterns between the data collected. The business can use customer insights to correlate purchasing behavior to demographics and enhance customer experiences through indirect monetization. Accept all reasonable answers.

3. A business notices they keep losing money during a specific quarter.

   The business should complete the data analytics process of examining raw data to draw conclusions and discover patterns. Focusing on diagnostic analytics will allow the company to investigate data to determine the causes of past events and understand why they are losing money. Accept all reasonable answers.

4. A business wants to improve their products, create new products, and increase employee happiness.

   First, the business should capture/collect data through transactions, surveys, and social media. Next, they should apply data correlation to customer insights and understand what products customers would like to see and what they would like to see updated. Apply operational efficiency to identify what impacts efficiency and quality to optimize workflow and avoid employee burnout. The business can also use indirect monetization to improve business operations, focusing on efficiency and product development. Accept all reasonable answers.

Putting It All Together (5-10 minutes)

Discussion Questions

- Review what students learned today by discussing the following questions as a class. Accept all reasonable answers.
- Why should companies use data to make business decisions?

  Companies should use data to make business decisions because it can give them the ability to make strategic decisions, gain competitive advantage over competition, and optimize improving processes and enhancing customer experiences.

- How can data monetization affect a business?

  Data monetization can help businesses use data to generate economic value. Businesses can sell data to other companies or use data to change their business processes expecting positive outcomes.

- How can businesses utilize data analytics?

  Businesses can utilize data analytics to discover patterns, correlations, and insights about their customers, products, past events, and business processes to inform the decisions they make and implement strategic actions.

- What is the difference between traditional data and big data?

  Big data is larger than traditional data in size, complexity, variety, and velocity. Big data has different processing tools and methods than traditional data solely because of its size.